



MD Rafiul Islam

Date of birth: 20 Nov 1999 | **Nationality:** Bangladeshi | **Phone number:** (+880) 1679899117 (Mobile) |

Email address: rafiulislam1921@gmail.com | **Website:** www.rafiulislam.me | **LinkedIn:** [rafi009](#) |

GitHub: [rafi0020](#) | **Google Scholar:** [Google Scholar](#)

ABOUT ME

I am a Machine Learning Engineer and AI researcher specializing in **Image Processing, Computer Vision, and Deep Learning**, with proven experience building **real-time, production-grade vision systems** (detection, tracking, OCR, pose-based reasoning, segmentation, and rule-based validation). My research focuses on **voice-to-spectrogram learning** for psychological stability analysis and has resulted in a Scopus indexed **Q1 journal publication**. I aim to advance my expertise in **image processing and computer vision** through the Erasmus Mundus IPCVAI programme and contribute to reliable, interpretable AI for healthcare and safety-critical applications.

WORK EXPERIENCE

MACHINE LEARNING ENGINEER – BONDSTEIN TECHNOLOGIES LTD. – 5 Feb 2025 – Current – DHAKA, BANGLADESH

Website: <https://www.bondstein.com>

- Lead delivery of **production-grade computer vision systems** from problem definition to deployment for enterprise clients.
- Design and maintain **real-time, multi-camera RTSP analytics pipelines** (low-latency inference, tracking, event logic).
- Build **end-to-end detection/OCR/pose-based modules** with strong post-processing and validation to reduce false alarms.
- Implement **temporal decision stability** (persistence, smoothing, hysteresis/state rules) and **evidence logging** for audit-ready alerts.
- Integrate AI outputs with **client APIs** and monitoring workflows; ensure reliability through **testing, logging, and restart-safe services**.
- Delivered systems using modular services (**watchers/senders**), persistent state management (**JSON/SQLite**), and **API integration** for scalable deployment and auditing.
- Collaborate with stakeholders to **refine requirements, evaluate performance**, and continuously **improve production accuracy** and robustness.

RESEARCH ASSISTANT – DAFFODIL INTERNATIONAL UNIVERSITY – 1 Jun 2024 – 3 Feb 2025 – DHAKA, BANGLADESH

Website: <http://research.daffodilvarsity.edu.bd/>

- Supported applied AI research across **signal processing, spectrogram-based learning, and computer vision**, including model development and research-grade evaluation design (**cross-validation, ROC-AUC** reporting).
- Designed and implemented deep learning models, including **Hybrid ViT-CNN** architectures, and contributed to research outputs in AI for healthcare.
- Supported dataset preparation, **augmentation**, class-imbalance handling (**SMOTE**), and **interpretability (XAI)** workflows such as **LIME, SHAP, and Grad-CAM** for research-grade experimentation.
- Published high-impact research findings in reputed journals, contributing to the academic discourse in **AI and healthcare**.
- Contributed to the creation of the first open-source Bengali voice dataset for mental health diagnostics. ([Repository](#))

EDUCATION AND TRAINING

18 JAN 2020 – 17 SEP 2024 Dhaka, Bangladesh

BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING Daffodil International University

Relevant coursework: Artificial Intelligence | Data Mining & Machine Learning | Natural Language Processing | Statistics & Probability | Numerical Methods | Operating Systems | Big Data & IoT | Human Computer Interaction

Undergraduate Thesis (Completed July 2024):

A NOVEL INTERACTIVE AI-BASED TOOL FOR DETECTING MENTAL STABILITY THROUGH ANALYSIS OF HUMAN VOICE

- Collected ethically Bengali voice recordings from mental health institutions in Bangladesh, with consent. The dataset includes 85 recordings across stable/unstable classes.
- spectrogram representations and applied hybrid CNN-Vision Transformer architectures for classification.
- Addressed severe class imbalance using SMOTE; implemented k-fold cross-validation.

Address Daffodil Smart City, Birulia, Dhaka, Bangladesh | **Website** www.daffodilvarsity.edu.bd | **Final grade** 3.52 out of 4.00 |

Level in EQF EQF level 6 | **Type of credits** Bangladeshi Credit System | **Number of credits** 148

● PUBLICATIONS

2024

[Mental Health Diagnosis from Voice Data Using Convolutional Neural Networks and Vision Transformers](#)

- Proposed a hybrid **ViT-CNN** approach using **log-mel spectrogram** representations of Bengali voice data.
- Achieved **91% accuracy** by leveraging CNNs for local patterns and Vision Transformers for long-range dependencies.
- Addressed limited real-world data with robust evaluation and class-balancing strategies, emphasizing clinical relevance and ethical data handling.

Islam, R., Ahad, M. T., Ahmed, F., Song, B., & Li, Y. (2024). Mental health diagnosis from voice data using convolutional neural networks and vision transformers. Journal of Voice. <https://doi.org/10.1016/j.jvoice.2024.10.010>

Authors: Islam, Rafiul; Ahad, Md Taimur; Ahmed, Faruk; Song, Bo; Li, Yan | **Journal Name:** Journal of Voice | **Publisher:** Elsevier BV

2025

[Using Computer Vision for Skin Disease Diagnosis in Bangladesh: Enhancing Interpretability and Transparency in Deep Learning Models for Skin Cancer Classification.](#)

- Computer vision benchmarking across multiple architectures with interpretability emphasis (XAI), targeting clinical transparency.

R Islam, JK Dipu, MH Tusar - arXiv preprint arXiv:2501.18161, 2025

Authors: Rafiul Islam, Jihad Khan Dipu, Mehedi Hasan Tusar | **Journal Name:** arXiv

● PROJECTS

MAR 2025

[Track My Container \(Industrial OCR + Validation System\)](#)

- Built a robust container number identification pipeline combining **geometric correction, detection, OCR, and domain validation**.
- Integrated OCR post-processing including **ISO check-digit validation** and correction heuristics to improve accuracy under real-world conditions.

[Live Demo](#)

MAR 2025

[AI SOP Checker \(Real-time SOP Compliance Auditing System\)](#)

- Designed and deployed a **multi-process, real-time computer vision auditing system** on live RTSP cameras to monitor factory SOP steps and generate evidence-based compliance events.
- Implemented modular checks (**bale opening + barcode scan, weighing step-back/unwanted activity detection, moisture inspection, layer-by-layer quality verification**) using YOLO-based models with **ROI/IoU-driven** logic.
- Engineered a **stateful reporting mechanism** using shared counters and per-check persistent state files to send first-time absolute counts and subsequent incremental updates reliably.

[Live Demo](#)

MAY 2025

[Bangla Automatic Number Plate Recognition \(ANPR\)](#)

- Designed a track-centric ANPR pipeline using YOLO-OBb detection, tracking, ROI-gated crop collection, and an OCR worker to finalize per-vehicle recognition.
- Implemented strict Bangladesh-format validation and voting-based plate selection, with cross-camera deduplication and rate limiting for production stability.

[Live Demo](#)

JUN 2025

[Argus Automata \(Warehouse Multi-camera Safety & Compliance Analytics\)](#)

- Built a multi-camera safety/security analytics platform covering **intrusion detection, overspeed estimation** (homography + Kalman smoothing), **collision-risk reasoning, PPE compliance** (helmet/vest), **safety violation detection** (temporal smoothing + zone logic), and **throwing detection**.
- Implemented stable alerting using tracking + temporal logic (EMA/hysteresis, dwell-time, cooldown) with **evidence snapshots** and **SQLite-backed alert queues** for downstream sending.

[Live Demo](#)

JUL 2025

[Staircase Safety Monitoring \(Pose-Based Reasoning\)](#)

- Implemented safety compliance reasoning using YOLO person detection + MediaPipe Pose landmarks.
- Used polygon rail ROIs and wrist/hand containment logic to detect handrail usage; detected phone usage via pose geometry; enforced temporal stability before logging and API reporting.

[Live Demo](#)

- Delivered a multi-module analytics platform: visitor flow, bike flow (in/out), bike model recognition, customer-product interaction detection, and service-delay analytics.
- Implemented employee vs customer desk classification with zone hysteresis, session persistence, identity logic, and decoupled uploader services.

[Live Demo](#)

- Built real-time crowd region detection using detection + tracking + monocular depth, with clustering and temporally smoothed heatmaps.
- Added weapon-aware escalation and suspect pinning logic using proximity and hysteresis-based persistence.

[Live Demo](#)

VOLUNTEERING

- Supported COVID-19 emergency response activities in a hospital vaccination programme.
- Assisted in patient information handling and operational support while maintaining privacy and confidentiality.

LEADERSHIP

Led club activities and contributed to student engagement initiatives.

HONOURS AND AWARDS

Discussed and presented "How the WIFI signals effects on the trees".

SKILLS

Soft Skill

Teamwork | Convincing | Inspiring | Influencing | Negotiation | Leadership skills

Digital Skills

Programming Languages: Python, C++, Java
Machine Learning / Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Transfer Learning, Data Augmentation, Explainable AI (LIME, SHAP, Grad-CAM)
Computer Vision Frameworks: OpenCV, Ultralytics YOLO, YOLO-OB, MediaPipe Pose, PaddleOCR, InsightFace, Depth-Anything-V2
Object Detection & Tracking: YOLO (v5/v8/v11), ByteTrack, DeepSORT, semantic segmentation (U-Net)
Audio & Signal Processing: Librosa, log-mel spectrogram extraction, noise-robust preprocessing, SMOTE for class balancing
Data & Experiment Tracking: Weights & Biases (W&B), MLflow (familiarity), TensorBoard
MLOps & Deployment: Docker, Jetson edge deployment (planning/optimization), TensorRT (FP16/INT8 quantization exploration), RTSP/GStreamer pipelines
Classical Image Processing: Geometric transforms (homography, fisheye undistortion), ROI masking, CLAHE contrast enhancement, morphological operations, temporal smoothing, hysteresis logic
Development Tools: Git, Linux, LaTeX, SQLite, JSON state management, multi-threaded/multi-process system design, API integration

LANGUAGE SKILLS

Mother tongue(s): **BENGALI**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	B2	C1	B2	B2	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user